INCH-POUND

MIL-DTL-45010C
23 April 2024
SUPERSEDING
MIL-DTL-45010B
02 September 2014

DETAIL SPECIFICATION

COMPOSITION C-4

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers four classes of Composition C-4, a military plastic explosive.

1.2 Classification. Composition C-4 is of the following classes:

Class 1 – with a plasticity of 0.030 units, maximum (max).
Class 2 – with a plasticity of 0.080 units, max
Class 3 – with a plasticity of 0.018 units, minimum (min).
Class 4 – with a plasticity of 0.030 units max and dyed.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4 and 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4 and 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards,and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

Comments box: Comments, suggestions, or questions on this document should be addressed to: U.S. Army DEVCOM Armaments Center, ATTN: FCDD-ACE-QSA, Picatinny Arsenal, New Jersey 07806-5000 or e-mailed to usarmy.pica.devcom-ac.list.ardec-stdzn-branch@army.mil. Since contact information can change, you may want to verify the currency of this information using ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 1376

Distribution Statement A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:12Z
Check the source to verify that this is the current version before use.

INTERNATIONAL STANDARDIZATION AGREEMENTS

AOP-7	-	Manual of Data Requirements and Tests for the Qualification of Explosive Materials for Military Use
STANAG 4170	-	Principles and Methodology for the Qualification of Explosive Materials for Military Use to

COMMERCIAL ITEM DESCRIPTIONS

A-A-59410	-	DMDNB (2, 3-Dimethyl-2, 3-Dinitrobutane)

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-398	-	RDX (Cyclotrimethylenetrinitramine)
MIL-P-14536	-	Polyisobutylene Binder

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-650	-	Explosive: Sampling, Inspection and Testing
MIL-STD-1168	-	Ammunition Lot Numbering and Ammunition Data Card
MIL-STD-1916	-	DOD Preferred Methods for Acceptance of Product

(Copies of these documents are available online at https://quicksearch.dla.mil/)

DEVCOM AC DRAWINGS AND QUALITY ASSURANCE PROVISIONS (QAPs)

7548645	-	Box, Fiberboard, Packing, Reusable, Collapsible, for High Explosives
7548645	-	Quality Assurance Provision for Box, Fiberboard, Packing, Reusable, Collapsible, for High Explosives
8833395	-	Charge, Demolition, Block, M5A1, Assembly

(Copies of these documents are available online at usarmy.pica.devcom-ac.list.drawing-request-help-desk@army.mil.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM E11	-	Standard Specifications for Woven Wire Test Sieve Cloth and Test Sieves
ASTM E300	-	Chemical, Industrial Sampling
ASTM D1193	-	Standard Specification for Reagent Water

(Copies of ASTM standards are available online from https://www.astm.org)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:12Z
Check the source to verify that this is the current version before use.

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2 and 6.13), first article inspection shall be performed in accordance with 4.2.

3.2 Conformance inspection. A sample shall be subject to conformance inspection in accordance with 4.3.

3.3 Material. Composition C-4 shall be comprised of RDX that is thoroughly and uniformly incorporated with a polyisobutylene binder to form a homogenous composition having a soft, putty-like consistency (see 6.2 and 6.10).

3.3.1 RDX. The RDX shall be in accordance with Type II RDX of MIL-DTL-398.

3.3.1.2 RDX granulation. The RDX content of Class 3 Composition C-4 shall consist of three parts coarse class, as specified in Table I, and one part nominal Class 5 RDX. All percentages shall be by weight using US Alternative Sieves conforming to ASTM E11.

TABLE I. Coarse class RDX granulation.

Through U.S. Alternative Sieve No.	Weight % passing
20	96-100
50	35-70
100	5-45
200	0-40

3.3.2 Polyisobutylene binder. The polyisobutylene binder shall be in accordance with Type I or Type II of MIL-P-14536 (see 6.11).

3.3.3 2,3-Dimethyl-2,3-dinitrobutane (DMDNB). The DMDNB shall be in accordance with A-A-59410.

3.4 Moisture. The moisture content shall be not greater than 0.25 percent.

3.5 Composition. The composition shall be as specified in Table II:

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:12Z
Check the source to verify that this is the current version before use.

TABLE II. Composition.

Class	Component	Weight Percent (%)
1 and 2	RDX	89.9 ± 1.0
	Polyisobutylene binder	8.9 ± 1.0
	DMDNB	1.2 ± 0.25
3	RDX	90.26 ± 0.7
	Polyisobutylene binder	9.49 ± 0.7
	DMDNB	0.25 ± 0.15/0.10
4 See 6.2 and 6.7	RDX	88.7 ± 1.0
	Polyisobutylene binder	9.9 ± 1.0
	DMDNB	1.2 ± 0.25
	Dye Content	0.2 ± 0.02
	Dye Composition	
	Lead Chromate	90
	Lamp Black	10

3.6 Insoluble particles. No particles shall be retained on a U.S. Alternative No. 40 sieve, and not more than five particles shall be retained on a U.S. Alternative No. 60 sieve from a 50 gram (g) portion of the sample. Sieves shall conform to ASTM E11.

3.7 Plasticity. The plasticity of Composition C-4 Class 1 shall be 0.030 units max. The plasticity of Composition C-4 Class 2 shall be 0.080 units max. The plasticity of Composition C-4 Class 3 shall be 0.018 units min. The plasticity of Composition C-4 Class 4 shall be 0.030 units max.

3.8 Specific gravity (applicable to Class 2 only). Composition C-4 shall have a specific gravity of 1.50 min, when pressed into shape of an M5A1 Demolition Block in accordance with Drawing 8833395. In the event that the manufacturer of the Composition C-4 also manufactures Composition C-4 demolition blocks, the specific gravity test may be waived and performed on the demolition block.

3.9 Large scale gap test (LSGT). The LSGT shall be conducted on the first article sample in accordance with 4.5.7. The LSGT data shall not be used to accept/reject the material.

3.10 Energetic materials. Composition C-4 and all constituent energetic materials used therein shall meet the requirements of AOP-7 and STANAG 4170 (see 6.14)

3.11 Workmanship. The Composition C-4 shall be free of foreign material such as dirt, rust, paint, metal chips, or any other foreign objects. Except for Class 4, Comp C-4 shall be off-white in color.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:12Z
Check the source to verify that this is the current version before use.

4. VERIFICATION

TABLE III. Requirement/verification cross reference matrix.

Method of Verification		Classes of Verification						
1 – Analysis 2 – Demonstration (end item test) 3 – Examination 4 – Test		A – First Article B – Conformance						
Section 3 and 5 Requirements	Description	Verification Method				Verification Class		Section 4
		1	2	3	4	A	B	
3.1	First article		X	X	X	X		4.2
3.2	Conformance inspection			X	X		X	4.4
3.3	Material			X		X	X	4.5.1
3.3.1	RDX			X		X	X	4.5.1.1
3.3.1.2	RDX granulation (for Class 3)			X		X	X	4.5.1.1.1
3.3.2	Polyisobutylene binder			X		X	X	4.5.1.2
3.3.3	DMDNB			X		X	X	4.5.1.3
3.4	Moisture				X	X	X	4.5.2
3.5	Composition	X	X		X	X	X	4.5.3
3.6	Insoluble particles				X	X	X	4.5.4
3.7	Plasticity				X	X	X	4.5.5
3.8	Specific gravity				X	X	X	4.5.6
3.9	LSGT				X	X		4.5.7
3.10	Energetic material			X		X		4.5.8
3.11	Workmanship			X		X	X	4.5.9
5.1	Packaging			X		X	X	4.5.10

4.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see paragraph 4.2)
b. Conformance inspection (see paragraph 4.3)

4.2 First article inspection. When specified, a sample shall be subjected to first article verification in accordance with Table IV. The testing shall apply for one (1) batch of Composition C-4 that has been produced by the contractor using the same production process, procedures, and equipment as will be used in regular production. All materials shall be obtained from the same sources as will be used in regular production.

4.3 First article rejection. If any item of the sample fails to comply with all first article requirements, the first article shall be rejected.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:12Z
Check the source to verify that this is the current version before use.

TABLE IV. First article acceptance tests.

	Composition C-4	Requirement Paragraph	Inspection Method
Critical:	None defined		
Major:			
101	Material	3.3	4.5.1
102	RDX	3.3.1	4.5.1.1
103	RDX granulation (for Class 3)	3.3.1.2	4.5.1.1.1
104	Polyisobutylene binder	3.3.2	4.5.1.2
105	DMDNB	3.3.3	4.5.1.3
106	Moisture	3.4	4.5.2
107	Composition	3.5	4.5.3
108	Insoluble particles	3.6	4.5.4
109	Plasticity	3.7	4.5.5
110	Specific gravity	3.8	4.5.6
111	LSGT	3.9	4.5.7
112	Energetic material	3.10	4.5.8
Minor:			
201	Workmanship	3.11	4.5.9
202	Packaging	5.1	4.5.10

4.4 Conformance inspection.

4.4.1 Lot formation. Lot formation shall be in accordance with MIL-STD-1916, Formation and identification of lots or batches. The criteria and procedures for the assignment of lot numbers shall be in accordance with MIL-STD-1168. A lot shall consist of one or more batches of C-4 not greater than 30 consecutive calendar days of production of one class designation only, produced by one manufacturer, in accordance with the same specification, or same specification revision, under one continuous set of operating conditions. Each batch shall consist of that quantity of C-4 that has been subjected to the same unit chemical or physical mixing process intended to make the final product homogeneous. In addition, each lot shall contain:

a. RDX of one type, from one lot interfix number and from one manufacturer only.
b. Polyisobutylene of one type, from one lot interfix number, and from one manufacturer only.

4.4.2 Sampling. The tests described in 4.5.1 to 4.5.9 shall be performed on samples representative of the batch which were taken in accordance with ASTM E300, for solids. Approximately 1 kg sample shall be taken for each batch for Classes 1, 3 and 4. Approximately 5 kg sample shall be taken for each batch for Class 2. Samples shall be stored in a plastic bag. If any sample fails to meet any test requirement, the batch represented by the sample shall be rejected.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:12Z
Check the source to verify that this is the current version before use.

4.4.3 Composition C-4				
Category	Examination or test	Verification level	Requirement paragraph	Inspection method
Critical	None			
Major				
101	Material	4.4.2	3.3	4.5.1
102	RDX	4.4.2	3.3.1	4.5.1.1
103	RDX granulation (for Class 3)	4.4.2	3.3.1.2	4.5.1.1.1
104	Polyisobutylene binder	4.4.2	3.3.2	4.5.1.2
105	DMDNB	4.4.2	3.3.3	4.5.1.3
106	Moisture	4.4.2	3.4	4.5.2
107	Composition	4.4.2	3.5	4.5.3
108	Insoluble particles	4.4.2	3.6	4.5.4
109	Plasticity	4.4.2	3.7	4.5.5
110	Specific gravity	4.4.2	3.8	4.5.6
111	LSGT	4.4.2	3.9	4.5.7
112	Energetic material	4.4.2	3.10	4.5.8
Minor		4.4.2		
201	Workmanship	4.4.2	3.11	4.5.9
202	Packaging	4.4.2	5.1	4.5.10

4.5 Test methods and procedures. (see 6.12)

4.5.1 Material. Verify visually/manually that the RDX is thoroughly and uniformly incorporated with a binder to form a homogenous composition having a soft, putty-like consistency.

4.5.1.1 RDX. Verify by examination of objective evidence that the RDX is in accordance with MIL-DTL-398.

4.5.1.1.1 RDX granulation. Verify by examination of objective evidence that Class 3 Composition C-4 consists of three parts coarse class, as specified in Table I, and one part nominal Class 5 RDX.

4.5.1.2 Polyisobutylene binder. Verify by examination of objective evidence that the polyisobutylene binder is in accordance with MIL-P-14536 (see 6.10).

4.5.1.3 DMDNB. Verify by examination of objective evidence that the DMDNB is in accordance with A-A-59410.

4.5.2 Moisture. The moisture shall be determined in accordance with MIL-STD-650, Method 101.4 or an approved equivalent method (see 6.9).

4.5.3 Determination of composition.

4.5.3.1 RDX content.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:12Z
Check the source to verify that this is the current version before use.

4.5.3.1.1 Titration method (non-aqueous). A portion of the sample accurately weighed to approximately 1.5 g and recorded to three decimal places shall be added to a 50 milliliter (mL) beaker. 25 mL of aliphatic naphtha shall be added and the beaker and contents transferred to a hot plate. The contents shall be swirled until all of the binder is dissolved as evidenced by the RDX crystals being separated. The beaker and contents shall be removed from the hot plate and cooled to room temperature. A fine porosity filter stick and suction shall be used to remove all of the aliphatic naphtha. 50 mL of dimethylformamide shall be added to the beaker and the mixture stirred until all of the RDX is dissolved. The solution shall be transferred quantitatively to a 100 mL volumetric flask and make up to volume with dimethylformamide. (A blank shall be run with dimethylformamide so that the blank is exposed to the air approximately the same amount of time as the sample.) A twenty (20) mL aliquot shall be withdrawn by pipette and transferred to a 100 mL tall form beaker containing 5 drops of azoviolet indicator solution and a magnetic stirrer.The beaker shall be covered with a glass cover containing a hole in the center to admit a buret tip and titrated with 0.1 normal sodium methoxide solution to a green endpoint which persists for 30 seconds. The magnetic stirrer shall be employed in the course of this titration. The titration procedure shall be repeated for the blank. The percentage of RDX in the sample shall be calculated as follows:

$$\text{Percentage RDX} = \frac{(A\text{-}B) \times (7.40) \times (N)}{W} \times 100$$

where:

A = volume of sodium methoxide used for the sample
B = volume of sodium methoxide used for the blank
N = normality of the sodium methoxide
W = weight of the sample on a moisture free basis represented by the aliquot taken

4.5.3.1.2 Gravimetric method. An accurately weighed portion of approximately 2 g shall be transferred to 100 mL beaker. Approximately 35 mL of reagent grade aliphatic naphtha saturated with RDX shall be added to the beaker. The beaker and contents shall be heated on a steam-heated hot plate. The beaker shall be covered and the contents stirred occasionally until the binder has gone into solution as evidenced by the RDX content being separated (approximately 30 minutes). The beaker and contents shall be removed from the steam hot plate and cooled to room temperature. The contents shall be filtered using a medium porosity glass crucible. The contents of the beaker shall be quantitatively transferred to the beaker crucible and washed with three 10 mL portions of reagent grade aliphatic naphtha saturated with RDX. The residue shall be aspirated until the reagent grade aliphatic naphtha has been completely removed. The crucible and contents shall be dried in steam oven maintained at 100 to 106°C for one hour or a steam heated vacuum drier for 20 minutes. The crucible shall be cooled in a desiccator for approximately 15 minutes and weighed. The percent RDX shall be calculated as follows:

$$\text{Percentage} = \frac{(A\text{-}B)}{W \times (1\text{-}M)} \times 100$$

where:

A = weight of crucible and residue
B = tared weight of crucible
M = percent moisture expressed as a decimal (see 4.5.2)
W = weight of sample

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:12Z
Check the source to verify that this is the current version before use.

4.5.3.2 Dye content. The manufacturing process shall be demonstrated to verify that the percent dye content has been mixed with composition in compliance with Table II.

4.5.3.3 DMDNB content. DMDNB content is determined via HPLC (High Performance Liquid Chromatography).

4.5.3.3.1 HPLC conditions.

a. Column –Reverse phase column yielding sufficient resolution and adequate retention time of analyte (e.g. Thermo Hypercarb; 3 micrometer particle size; 50 mm x 4.6 mm I.D.)

b. Mobile phase –50% (by volume) distilled, deionized water, 4% (by volume) HPLC grade acetonitrile; 46% (by volume) HPLC grade methanol.

c. Flow rate –1.15 mL/min.

d. Injection volume –7 µL.

e. Temperature - 30ºC

f. Run time – 10 minutes.

Note - HPLC parameters (e.g. flow rates, mobile phase ratios, injection volumes, run times) may be adjusted to maintain adequate resolution and shape of the DMDNB peak and any impurities if present. Elution order is DMDNB, RDX, and HMX.

4.5.3.3.2 Standard preparation. Using analytically pure DMDNB (>99.5% purity), prepare 5 standards with the approximate concentrations shown below:

Table V. Standard preparation.

STD	Concentration (g/mL)	Target Weight (g)	Dilution volume (mL) w/ HPLC grade Methanol
1	0.00010	0.0100	100
2	0.00015	0.0150	100
3	0.00020	0.0200	100
4	0.00030	0.0300	100
5	0.00035	0.0350	100

4.5.3.3.3 Sample preparation. Prepare four individual samples by weighing approximately 2 g of each sample, recorded to four decimal places, and diluting to 25 ml with HPLC grade methanol. This will yield samples with an approximate DMDNB concentration of 0.00020 g/mL.

4.5.3.3.4 Analysis by HPLC & calculation of results. Once the HPLC system has been equilibrated, run all standards and samples to obtain peak areas for each. Using an appropriate software package (e.g., MS Excel), establish a calibration equation based on the concentrations of DMDNB in the five calibration standards and the HPLC chromatographic peak areas obtained

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:12Z
Check the source to verify that this is the current version before use.

from the analytical testing of the standards. The software utilizes a least squares regression analysis to determine the equation of fit for a straight-line regression. Using the calculated equation, the concentration of DMDNB in the samples can be determined.

Report the average of all four replicates as the % DMDNB for the batch.

4.5.3.4 Polyisobutylene binder. The percent of polyisobutylene binder shall be calculated on a dry basis by subtracting from 100 percent the percent of RDX as obtained in 4.5.3.1 and the percent of DMDNB as obtained in 4.5.3.3 for Composition C-4 Classes 1, 2, and 3. For Composition Class 4, the percent of polyisobutylene binder shall be calculated on a dry basis by subtracting from 100 percent the percent of RDX as obtained in 4.5.3.1, the percent ofDMDNB as obtained in 4.5.3.3, and the percent of Dye content per 4.5.3.2.

4.5.4 Determination of insoluble particles. A weighed portion of approximately 50 g of the sample shall be placed in a 600 mL beaker. A 400 mL aliquot of petroleum ether or reagent grade aliphatic naphtha shall be added and the sample heated on a steam bath until all lumps are broken down and all soluble material is dissolved. The solution shall be decanted through a number 40 U.S. Standard sieve placed on a number 60 U.S. Standard sieve. The insoluble material shall be retained in the beaker. Acetone shall be added to the beaker and the beaker and contents warmed on a steam bath to dissolve the insoluble matter. This mixture shall be poured through the nest of sieves making sure all insoluble matter is transferred to the sieves. Any residue left on the sieve shall be washed with acetone. The sieves shall be dried on a closed steam bath, and the particles of residue counted.

4.5.5 Plasticity. The plasticity shall be determined in accordance with MIL-STD-650 method 211.2.

4.5.6 Specific gravity.

4.5.6.1 Molds. Any suitable mold may be used which will permit 2.50 ± 0.5 pounds of Composition C-4 to be pressed with a 5 ton load to form a block approximately 2 inches thick and the area of Composition C-4 supporting the load to be 22 square inches.

4.5.6.2 Preparation for determination of specific gravity. The mold, except for top cover, shall be assembled and a sheet of 5 to 10 mils thick cellophane or wax paper, cut to size, placed on the inside of the assembled mold. A weighed charge of 2.50 ± 0.5 pounds of Composition C-4 shall be placed in the mold and manually distributed uniformly in the mold. The ends of the cellophane or wax paper shall be folded over the charge. The top cover of the mold shall be placed over the charge. The loaded assembly shall be transferred to a hydraulic press and the charge pressed using a load of five tons. A five-second dwell shall be used, the press retracted and the pressing operation repeated once. The block shall be removed from the mold and the cellophane or wax paper shall be carefully removed from the pressed block. Any excess explosive shall be trimmed from the edges of the block. The block shall be stored at 25 ±2°C for 15 minutes on a flat surface.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:12Z
Check the source to verify that this is the current version before use.

4.5.6.3 Determination of specific gravity. The block shall be weighed to the nearest 0.01 pound when dry. The block shall be weighed immersed in water with water temperature at 77 ± 9°F (25 ± 5°C), taking care to remove all air bubbles remaining on the surfaces of the block. The water temperature shall be recorded. The specific gravity shall be calculated as follows:

$$\text{Specific Gravity} = \frac{A}{A-B} \times C$$

where

A = weight of block dry
B = weight of block immersed in water
C = specific gravity of water at recorded temperature

4.5.7 Large Scale Gap Test (LSGT).

a. Gap test and configuration shall be performed in accordance with United States' procedures in AOP-7, US 201.04.002 for the LSGT (NOL).

b. Individual pellet density range shall be 1.600 ± 0.020 g/cc. Overall charge density shall be 1.600 ± 0.020 g/cc.

c. In the LSGT test, a positive result or "go" is recorded when a neat hole is punched in the plate. A broken plate or one with a poor quality hole is considered a "no-go." Report result as the card gap thickness that gives a 50% probability of detonation.

4.5.8. Energetic material. Verify by visual examination of objective evidence that Composition C-4 and all constituent energetic materials meet the requirements of AOP-7 and STANAG 4170 (see 6.14).

4.5.9 Workmanship. Composition C-4 shall be visually inspected for foreign material such as dirt, rust, paint, metal chips or any other foreign objects. For Classes 1, 2 and 3 verify by visual examination that the Composition C-4 is off white in color.

4.5.10 Packaging. Composition C-4 in packaged configuration shall be sampled and inspected in accordance with Quality Assurance Provisions of QAP 7548645 during first article and lot acceptance testing. Any defect shall reject the lot.

5. PACKAGING

5.1 Packaging. Composition C-4 shall be packaged in accordance with Drawing 7548645.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:12Z
Check the source to verify that this is the current version before use.

6.1 Intended use. The material covered in this specification is intended to be used as an extruded or pressed explosive for application where high energy and ability to be shaped by hand are necessary.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Ammunition Data Cards (see 6.3)

b. Propellant and Explosive (P&E) Data Sheets (see 6.4)

c. Title, number and date of this specification.

d. Requirements for submission of first article sample.

e. Special marking, when specified.

f. Certificate of conformance for each lot or shipment of product (if applicable).

g. Government Furnished Material or equipment (if applicable).

h. Process Control Document (see 6.13 if applicable).

6.3 Ammunition data cards. When specified in the contract or purchase order (see 6.2), one data card prepared in accordance with MIL-STD-1168 should be included with each container (see 5.1). If the information contained on the data card is provided with product acceptance documents and the Composition C-4 batch numbers and net weight of contents are stenciled on the shipping containers, then data cards should not be required.

6.4 Contract data requirements. When stated in the contract, acceptance and description sheets should be prepared by the contractor to show compliance with the specifications provided in section 3. The Worldwide Ammunition-data Repository Program (WARP) should be utilized to store the data sheets required by MIL-STD-1171A. The Munitions History network located at https://mhp.redstone.army.mil/ should be used to gain access to WARP.

6.5 Equivalent test methods. The test methods given in this specification are the required methods to be used. The contractor may request using other methods providing that the proposed method is equivalent (accuracy and precision) to the method given in this specification. Prior approval of the Contracting Officer is required for use of equivalent test methods. A description of the proposed method should be submitted through the Contracting Officer to the technical agency. This description should include, but not be limited to, the detailed test procedures, required equipment and materials, accuracy and precision of the method, test data to demonstrate the accuracy and precision and Drawings of any special equipment required.

6.6 Contractor acceptance inspection equipment (AIE). Provisions concerning the contractor's AIE used to verify the requirements of this specification should be specified in the contract.

6.7 Submission of alternative quality conformance provisions. All contractor proposed alternative quality conformance provisions should be submitted to the Government for evaluation/approval by the technical agency responsible for the preparation of this specification.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:12Z
Check the source to verify that this is the current version before use.

6.8 Method of manufacture for Dyed Composition C-4. The lead chromate and lamp black should be mixed thoroughly before adding to the polyisobutylene at a rate of 4 to 5 grams per minute. The mixture should be agitated slowly in a folding action at 90°C. After all the pigment has been added, the mixture should be added to an RDX/water slurry at 21°C with fast agitation. Then the composition C-4 should be poured directly into a vacuum pan to draw off the water. (At this point the C-4 should have the appearance of tapioca pudding). The composition should be processed in a Wabash type Incorporator for 30 minutes and then dried in a drying house.

6.9 Technical agency. DEVCOM AC is the technical agency that is responsible for the preparation and revision of the specification. All correspondence to the Technical Agency should be submitted through the contracting officer to: DEVCOM AC, ATTN: FCDD-ACE-QEE, Picatinny Arsenal, NJ 07806-5000.

6.10 Moisture determination. Holston AAP Analytical Standard Method, ASM I-7 is approved as an equivalent method and may be used in lieu of Method 101.4 of MIL-STD-650.

6.11 Holston slurry process. The alternate method of manufacture of Composition C-4 is by the Holston Slurry Process (MI No 1131-1208 for Lacquer Preparation C-4 and MI No 1143-1201 RDX Coating - Comp C-4). Manufacture of Composition C-4 by the Holston Slurry Process exempts the manufacturer from the requirements of MIL-P-14536 except as follows:

a. Paragraph 1 and 2 should remain in effect and in addition, the manufacture should use only raw materials conforming to military specification cited in paragraph 2.

b. The ratio of binder components should be in accordance with paragraph 3.1.3 and should be controlled on the basis of input quantities to the Holston Slurry Process. MIL-P-14536 should be adhered to entirely when any other Composition C-4 manufacturing process is used or when the polyisobutylene binder exists as an isolable intermediate.

c. Di-(2-ethylhexyl)-adipate is the qualified constituent used in the manufacturing of polyisobutylene binder (PIB) for Holston slurry process.

6.12 Composition C-4. Composition C-4 manufactured before January 1994 and in accordance with MIL-C-45010A (Amendment 4) does not contain DMDNB in the formulation of Composition C-4.

6.13 Hazardous materials. This specification covers sampling and testing of toxic or hazardous materials. Accordingly, it is emphasized that all applicable safety rules, regulations and procedures should be followed in handling and processing these materials.

6.14 Process control document. When a contract is awarded, a Process Control Document should be submitted to the Technical Agency (see 6.8) prior to the First Article Test. This document should contain a description of the process, all materials used, process conditions/procedures, and production/inspection equipment used to produce Composition C-4 meeting the requirements of this specification. In addition, whenever there is a change that requires lot interfix change (per MIL-STD-1168) such as the source of material, process conditions, procedures, etc. from those in the previous process control document, the Technical Agency (see 6.8) should be notified of the new process control change(s) and the reason for the change(s) within 10 business days from the change.

6.15 Energetic material approval. Qualification of energetic materials is a demonstration

Source: https://assist.dla.mil -- Downloaded: 2023-12-21 02:42
Check the source to verify that this is the current version before use.

that the energetic material produced at one facility by one manufacturing process meets the requirements of AOP-7 and STANAG 4170. The qualification is limited to one intended military use, e.g. qualification as a primary explosive does not also mean it is qualified as a main charge explosive. Point of contact for the U.S. Army Qualification Authority, a list of approved supply sources, and/or test matrix outlining the requirements of AOP-7 and STANAG 4170 may be obtained from the Technical Agency (see 6.8).

6.16 Incorporated changes. ECP R19Q2009 and R22Q5003 provide documentation for noted changes.

6.17. Subject term (key word) listing.

Plastic explosive
Moldable explosive
Demolition Charge
Polyisobutylene
M112
MICLIC
Claymore
PIB
RDX

6.18 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodian:
Army-AR
Navy-OS
Air Force- 99

Preparing Activity:
Army-AR
(Project 1376- 2024-004)

Review activities:
Army-AV,TE
Navy-SH
Air Force-11, 70

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:12Z
Check the source to verify that this is the current version before use.